Bahria University

Discovering Knowledge

FYP PROPOSAL DEFENSE PROFORMA

2025-26

Title of the Project

SalesChamp: An AI-Powered Multi-Agent Sales Automation Platform with Salesforce CRM, Microsoft Dynamics 365 Sales, and SAP ERP Integration

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Introduction

Enterprise sales operations sit at the intersection of two disciplines that have historically been handled by separate systems: customer relationship management, which tracks who a business is talking to, and enterprise resource planning, which tracks what a business can actually deliver. Over the last two years, the rapid maturity of Large Language Models (LLMs), Retrieval Augmented Generation (RAG) and agentic AI frameworks such as LangGraph and Google's Agent Development Kit has opened a practical path toward software that does not merely store sales data but reasons over it, plans multi step workflows and executes them with a human reviewer in the loop.

This project sits in the domain of Agentic AI and enterprise software integration. The motivation for choosing this domain comes from a simple observation: modern CRM platforms such as Salesforce and Microsoft Dynamics 365 Sales are excellent repositories of customer information, and modern ERP platforms such as SAP are excellent systems of record for inventory, quotations and invoicing, yet these systems rarely talk to each other in real time or across CRM ecosystems, and none of them performs the research, drafting and follow up work that a sales representative still has to do by hand.

The existing gap for innovation, therefore, is not in building another CRM dashboard, but in building an orchestration layer of cooperating AI agents that can discover leads, research them, draft tailored proposals, keep Salesforce, Microsoft Dynamics 365 Sales and SAP synchronised regardless of which CRM an organisation has adopted, and escalate only the decisions that genuinely require a human, so that sales representatives spend their time closing deals rather than updating records.

Problem Statement

Problem Context

Sales representatives in small and mid sized technology and services firms typically divide their week between prospecting, research, proposal writing, CRM data entry and coordination with finance or inventory teams. Industry surveys on sales productivity consistently report that administrative and non selling activity consumes a large share of a representative's working time, which directly reduces the number of qualified conversations a team can hold in a given month.

The problem is compounded when a business uses a CRM and an ERP as disconnected systems, or when different parts of the same organisation standardise on different CRMs. A deal that has been won in Salesforce or Microsoft Dynamics 365 Sales still has to be re entered into SAP as a customer master record, a sales order and, eventually, an invoice, and a customer tracked in one CRM often has no visibility in the other. Every manual re entry is a point where data can drift out of sync, and every delay between systems is a delay in delivering the customer's order.

Current Challenges

- Manual lead discovery through directories, maps and company websites, which does not scale beyond a handful of representatives
- Time consuming, largely manual research into a prospect's business model, technology stack and likely pain points
- Proposal documents drafted from scratch for each prospect, with inconsistent quality and turnaround time
- CRM records that fall behind actual conversations because updates depend on a representative remembering to log them
- No automated bridge between a won opportunity in the CRM and the quotation, sales order and invoice lifecycle in the ERP
- Follow up communication that is either forgotten or handled inconsistently across the sales team
- Customer and pipeline information trapped inside a single CRM ecosystem, with no automated way to keep Salesforce and Microsoft Dynamics 365 Sales records consistent when an organisation runs both

Client / Stakeholder Perspective

The primary stakeholders are sales representatives and sales managers at organisations that run Salesforce or Microsoft Dynamics 365 Sales for customer relationship management and SAP for enterprise resource planning, a combination common among mid sized manufacturing, distribution and technology firms, including organisations where different business units standardise on different CRMs. For these teams, the cost of the problem is measured directly in lost selling hours and in the lag between winning a deal and fulfilling it, both of which SalesChamp is designed to reduce.

Impact of the Problem

The absence of an integrated, agent driven sales platform carries consequences across several dimensions of the business:

- **Technical Impact:** Sales data drifts out of sync between Salesforce, Microsoft Dynamics 365 Sales and SAP, producing duplicate customer records, inaccurate inventory checks and quotations that do not reflect current stock.
- **Operational Impact:** Representatives spend a disproportionate share of their week on prospecting research, data entry and manual follow up rather than on direct selling activity.

- **Economic Impact:** Slow, inconsistent proposal turnaround and delayed handoffs between the CRM and the ERP extend the overall sales cycle and raise the effective cost of closing each deal.
- **Client Impact:** Customers experience longer gaps between agreeing a deal and receiving their quotation, sales order or invoice, which can affect satisfaction and the likelihood of repeat business.

Proposed Solution

Project Features

SalesChamp is proposed as a multi agent sales automation platform that sits alongside an organisation's existing Salesforce, Microsoft Dynamics 365 Sales and SAP deployments rather than replacing them. Instead of a single chatbot answering questions, the system is composed of specialised agents, each responsible for one stage of the sales lifecycle, coordinated by a Manager Agent that decides which agent acts next and when a human approval step is required. Because the platform supports both Salesforce and Microsoft Dynamics 365 Sales, it remains CRM independent and can adapt to whichever enterprise environment a customer already runs, without changing the underlying AI workflow.

Key Features

- **AI Lead Discovery**
  - Identifies prospective businesses through Google Maps, public business directories and company websites, ranking them by estimated sales potential.
- **Intelligent Business Research**
  - Compiles a profile of each prospect covering products, competitors, digital presence, technology stack and likely pain points, stored in a shared knowledge base for reuse by other agents.
- **Automated Proposal Generation**
  - Drafts a proposal customised to the prospect's stated requirements using Retrieval Augmented Generation over the organisation's own product and pricing material.
- **Multi-CRM Synchronisation (Salesforce & Microsoft Dynamics 365 Sales)**
  - Creates leads, contacts and opportunities in Salesforce or Microsoft Dynamics 365 Sales, logs every interaction automatically, synchronises customer records bi-directionally between the two platforms and removes duplicates, so representatives can keep working from whichever CRM their organisation uses while the platform handles routine record keeping.
- **SAP ERP Integration**

    - Once a lead is converted, the platform creates the customer master record, generates quotations, checks inventory and raises sales orders and invoices without duplicate data entry.
- **AI Voice and Email Follow Up**
    - Schedules and places reminder communications with prospects who have gone quiet, logging the outcome back into the CRM.
- **Human in the Loop Approval**
    - Routes sensitive actions, such as sending a quotation or creating a SAP sales order, to a human representative for confirmation before they take effect.
- **Executive Analytics Dashboard**
    - Summarises pipeline health, conversion rates and forecasted revenue for sales managers.
- **Microsoft Copilot and Power Platform Readiness**
    - Structures Dynamics 365 activity so it can trigger Power Automate workflows and surface Microsoft Copilot recommendations, positioning the platform for deeper Microsoft ecosystem integration as a future scope item.

Methodology / Algorithm

The platform is organised as a pipeline of cooperating agents rather than a single monolithic model. A Manager Agent receives each new task, be it a fresh lead, an incoming email reply or a scheduled follow up, and delegates it to the appropriate specialised agent, waiting for that agent's result before deciding on the next step. This design keeps each agent's responsibility narrow and testable, and lets the team add or replace an agent, for example swapping the SAP Agent's integration method, without disturbing the rest of the pipeline.

Eight specialised agents report to the Manager Agent: a Lead Discovery Agent, a Research Agent, a Proposal Agent, a multi-CRM Agent, an SAP Agent, a Communication Agent, a Follow up Agent and an Analytics Agent. The CRM Agent is the only component permitted to write to Salesforce or Microsoft Dynamics 365 Sales, and the SAP Agent is the only component permitted to write to SAP, which keeps external side effects contained to two well defined integration points even though the CRM Agent may address either platform.

Algorithm

1. Receive a new task, such as a fresh lead, an incoming email reply or a scheduled follow up, at the Manager Agent.
2. Delegate the task to the Lead Discovery Agent to identify and rank a candidate business.

3. Pass the candidate to the Research Agent to compile a profile covering products, competitors and technology stack.
4. Forward the profile to the Proposal Agent to draft a tailored proposal using Retrieval Augmented Generation.
5. Route the draft proposal to a human representative for review and approval under the Human in the Loop policy.
6. On approval, trigger the Communication Agent to send the proposal and the CRM Agent to create the lead in whichever CRM, Salesforce or Microsoft Dynamics 365 Sales, the organisation has adopted.
7. When a lead responds positively and is marked qualified, direct the CRM Agent to create and synchronise the opportunity across CRMs where applicable, then hand off to the SAP Agent.
8. Use the SAP Agent to generate a quotation, verify inventory and raise a sales order, followed by an invoice once payment is confirmed.
9. Update the executive analytics dashboard through the Analytics Agent at every step of the flow.

Architecture / Workflow Diagram

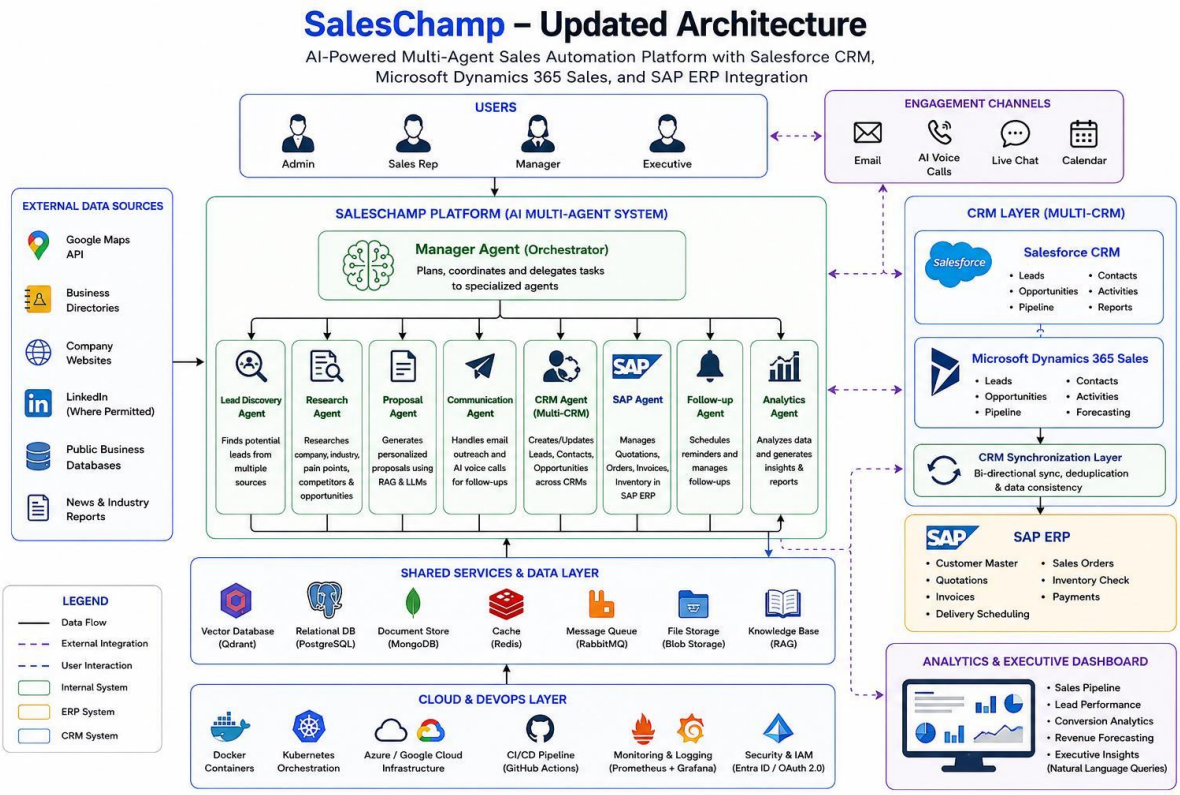


Fig 1: Updated SalesChamp architecture, showing the Manager Agent coordinating the Lead Discovery, Research, Proposal, Communication, multi-CRM, SAP, Follow-up and Analytics agents across the shared services layer, the Salesforce and Microsoft Dynamics 365 Sales CRM layer, and the SAP ERP layer.

Technologies to Be Used

The technology choices below favour actively maintained, well documented frameworks so that the final year implementation can rely on stable APIs for both the AI orchestration layer and the enterprise integrations.

Layer	Technologies
Frontend	React.js or Next.js with Tailwind CSS
Backend	Django REST Framework or FastAPI
Agent Orchestration	LangGraph, with Google's Agent Development Kit (ADK) as an alternative orchestration layer
Language Models	GPT-5, Gemini and Azure OpenAI Service, selected per task through a model router
Vector Database	ChromaDB for the campus prototype, with Qdrant as the production scale option
Relational Database	PostgreSQL
CRM Integration	Salesforce REST and Apex APIs; Microsoft Dynamics 365 Web API, Microsoft Dataverse and Microsoft Graph API, authenticated through OAuth 2.0
ERP Integration	SAP Business Technology Platform (BTP) APIs and OData services; Microsoft Power Platform (Power Automate) for future Copilot-assisted workflows
Authentication	OAuth 2.0 and JSON Web Tokens (JWT)
Deployment	Docker containers, with Kubernetes considered for the production scale future scope

Innovation of the Proposed Solution

Unlike conventional CRM add ons or single purpose ERP connectors that operate within a single ecosystem, SalesChamp brings agentic AI orchestration, Retrieval Augmented Generation, human in the loop approval and synchronisation across Salesforce, Microsoft Dynamics 365 Sales and SAP together in one coordinated, CRM-independent platform. Rather than automating a single task in isolation, the system lets a representative move from lead discovery through research, proposal drafting, multi-CRM logging and ERP fulfilment within one continuous, supervised workflow, so the value of automation compounds across the whole sales cycle and across heterogeneous enterprise systems instead of being confined to one step, or one CRM, at a time.

Sustainable Development Goals

SalesChamp contributes most directly to two of the United Nations Sustainable Development Goals, summarised below with the corresponding justification.

SDG	Relevance to the Project
SDG 8	Decent Work and Economic Growth: by automating repetitive administrative work, the platform lets sales professionals redirect their effort toward higher value, more meaningful selling activity, and it helps small and mid sized enterprises compete for business that would otherwise require a larger sales headcount.
SDG 9	Industry, Innovation and Infrastructure: the project demonstrates a practical, exportable pattern for connecting agentic AI to existing enterprise infrastructure such as Salesforce and SAP, contributing to the broader modernisation of business process infrastructure rather than requiring organisations to replace systems they already depend on.

Work Breakdown Structure / Gantt Chart

The project is planned across the two Final Year Project semesters, with FYP-I covering requirements, design and the first integration milestones, and FYP-II covering full agent development, ERP synchronisation, testing and the final defence. The phases below decompose the project into manageable, testable units of work, and the tentative task distribution over time is shown in the Gantt chart that follows.

Gantt Chart

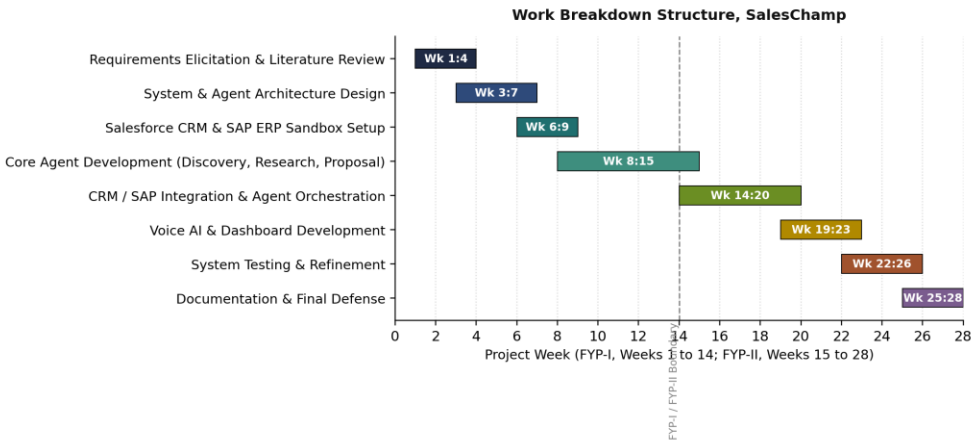


Fig 2: Tentative Gantt chart for SalesChamp across FYP-I and FYP-II.

Work Breakdown Structure

10. Requirements Elicitation and Literature Review: confirming scope with the supervisor, studying existing agentic sales tooling and finalising the feature set.

11. System and Agent Architecture Design: defining the Manager Agent's coordination logic and the responsibilities of each specialised agent.
12. Salesforce and SAP Sandbox Setup: provisioning developer accounts and confirming API access ahead of integration work.
13. Core Agent Development: implementing the Lead Discovery, Research and Proposal agents against sandbox and public data.
14. CRM and SAP Integration: connecting the CRM and SAP agents to live sandbox environments and validating two way data consistency.
15. Voice AI and Dashboard Development: building the follow up communication layer and the executive analytics dashboard.
16. System Testing and Refinement: end to end testing of the pipeline, including failure and approval override scenarios.
17. Documentation and Final Defence: preparing the final report, demonstration and defence presentation.

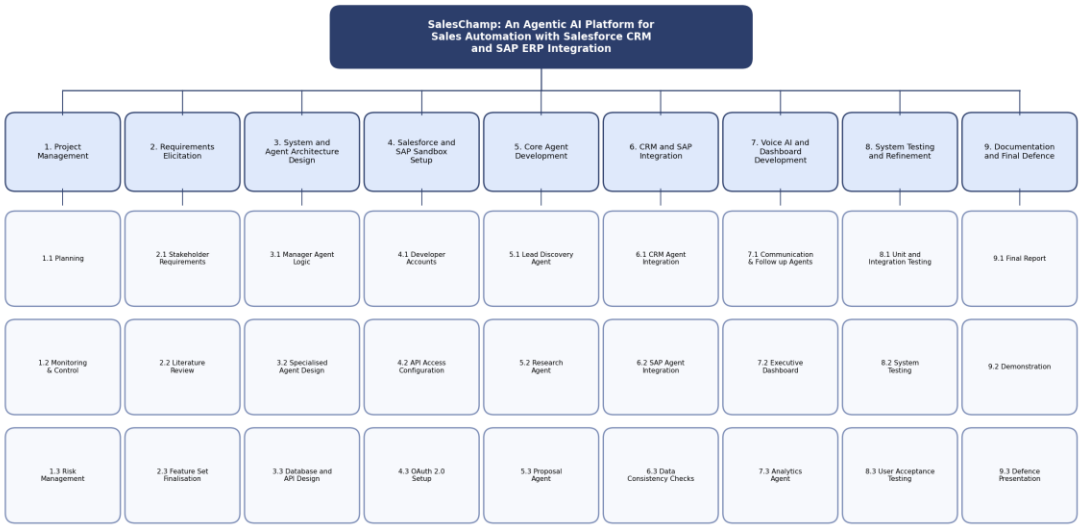


Fig 3: Work breakdown structure for SalesChamp across FYP-I and FYP-II.

References

[1] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” in Proc. Advances in Neural Information Processing Systems (NeurIPS), 2020, pp. 9459 to 9474.

[2] Y. Wu et al., “Autonomous Agents Based on Large Language Models: A Survey,” arXiv preprint arXiv:2308.11432, 2023.

[3] LangChain AI, “LangGraph Documentation,” 2024. Available: https://langchain-ai.github.io/langgraph/

[4] Google, “Agent Development Kit Documentation,” Google Developers, 2025. Available: https://google.github.io/adk-docs/

[5] Salesforce, “REST API Developer Guide,” Salesforce Developers Documentation, 2025. Available: https://developer.salesforce.com/docs/

[6] SAP SE, “SAP Business Technology Platform, OData Services Overview,” SAP Help Portal, 2025. Available: https://help.sap.com/

[7] OpenAI, “OAuth 2.0 and API Authentication Best Practices,” OpenAI Platform Documentation, 2025. Available: https://platform.openai.com/docs/

[8] S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4th ed. Harlow, U.K.: Pearson, 2021.

[9] M. Wooldridge, An Introduction to MultiAgent Systems, 2nd ed. Chichester, U.K.: John Wiley and Sons, 2009.

SUPERVISOR CONSENT FORM
(DEPARTMENT OF SOFTWARE ENGINEERING)

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Group Members:

Enrollment #	Name
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02-131232-079	Umer Hussain (Team Lead)
02-131232-121	Salman Khan

Supervisor Declaration:

☐ I hereby confirm that I have reviewed the project proposal document and agree to supervise the project.

Supervisor: _____

Signature: _____

Date: _____

FYP Coordinator:

_____

Signature:

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Date:

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